Mississippi Valley State University

CS 341 — Discrete Structures

Lecture 7: Always True, Always False

Tautologies, Contradictions, and Two Rules of Inference

Chartrand & Zhang, Section 1.5

Name: _____ Date: _____

Where we are

Every compound statement you have built so far has been true in some rows and false in others. Today we look at the two extreme cases — always true and always false — and find that the "always true" ones are exactly the rules of reasoning you will use in every proof from Chapter 3 onward. This closes Chapter 1.

By the end of today you should be able to

1. Decide whether a compound statement is a tautology, a contradiction, or neither.
2. State and verify modus ponens and modus tollens.
3. Recognise the two classic fallacies that look like them and are not.

1. Two extremes

Definitions 1.59 and 1.60

A compound statement is a **tautology** if it is true for *all* combinations of truth values of its component statements.
A compound statement is a **contradiction** if it is false for *all* combinations.

The simplest examples use a single component statement.

P	$\sim P$	$P \vee (\sim P)$	$P \wedge (\sim P)$
T	F	T	F
F	T	T	F

$P \vee (\sim P)$ is a tautology; $P \wedge (\sim P)$ is a contradiction.

Two facts that follow immediately

(1) S is a tautology if and only if $\sim S$ is a contradiction.
(2) If $R \equiv S$, then $R \Leftrightarrow S$ is a tautology. Every equivalence you proved in Weeks 1 and 2 was secretly a tautology.

With an open sentence. Let $P(n)$: n is even, over the integers.

- $P(n) \vee (\sim P(n))$: n is even or n is odd — true for every integer.
- $P(n) \wedge (\sim P(n))$: n is even and n is odd — false for every integer.

2. Classify these

P	Q	$P \Rightarrow Q$	$\sim Q$	$(\sim Q) \vee (P \Rightarrow Q)$	$(P \wedge Q) \wedge (Q \Rightarrow (\sim P))$
T	T	T	F	T	F
T	F	F	T	T	F
F	T	T	F	T	F
F	F	T	T	T	F

Column five is a tautology. Column six is a contradiction.

3. Modus ponens — the rule you already use

Modus ponens

$$(P \wedge (P \Rightarrow Q)) \Rightarrow Q$$

If P is true, and P implies Q, then Q is true.

P	Q	$P \Rightarrow Q$	$P \wedge (P \Rightarrow Q)$	$(P \wedge (P \Rightarrow Q)) \Rightarrow Q$
T	T	T	T	T
T	F	F	F	T
F	T	T	F	T
F	F	T	F	T

True in every row, so it is a tautology.

4. Modus tollens — the contrapositive at work

Modus tollens

$$((P \Rightarrow Q) \wedge (\sim Q)) \Rightarrow (\sim P)$$

If P implies Q, and Q is false, then P is false.

This is also a tautology — verify it on your own paper.

You have already used both

Quiz 2 asked: the requirement "if a file is encrypted, then it is backed up" holds, and a file is *not* backed up. Concluding the file is not encrypted is exactly modus tollens.
These two tautologies are the engine of every proof in Chapter 3. Modus ponens drives direct proof; modus tollens drives proof by contrapositive.

5. Two impostors

Change one symbol in each rule and it stops being a tautology.

P	Q	$(Q \wedge (P \Rightarrow Q)) \Rightarrow P$	$((P \Rightarrow Q) \wedge (\sim P)) \Rightarrow (\sim Q)$
T	T	T	T
T	F	T	T
F	T	F	F
F	F	T	T

Neither is a tautology. Both fail in the same row: P false, Q true.

These have names, and you have met them before

The first is **affirming the consequent**: knowing Q and $P \Rightarrow Q$, concluding P. That is assuming the *converse*.
The second is **denying the antecedent**: knowing $\sim P$ and $P \Rightarrow Q$, concluding $\sim Q$. That is assuming the *inverse*.
Week 2 showed converse and inverse are equivalent to each other and *not* to the original. That is precisely why both fallacies fail, and why they fail in the same row.

Which rule, if any? Mark each argument **valid** or **fallacy**.

Argument	Verdict
If the tests fail, the build stops. The tests failed. So the build stopped.	Valid — modus ponens
If the tests fail, the build stops. The build stopped. So the tests failed.	Fallacy — converse
If the tests fail, the build stops. The build did not stop. So the tests did not fail.	Valid — modus tollens

Exit ticket

1. Is $(P \Rightarrow Q) \vee (Q \Rightarrow P)$ a tautology, a contradiction, or neither? One line of justification.

2. Name the rule: "If it rained, the ground is wet. The ground is not wet. So it did not rain."

Before Wednesday

Read *Mathematics for Computer Science* (Lehman, Leighton, and Meyer), Section 3.4.1, "Propositions in Normal Form," pp. 50–52. The link is on Canvas.

Note: Chartrand does not cover normal forms, so Wednesday and Friday use the free supplement instead. Everything else this term stays in Chartrand.

Self-check (not collected). Section 1.5, Exercises 1, 3, 5, 7, and 9 — all odd-numbered, so answers are in the back. Exercise 5 is modus tollens.

Coming up

Homework 3 is due Friday, September 10, 11:59 p.m.
Wednesday and Friday cover conjunctive and disjunctive normal form — the last topics in our logic unit.

Questions I still have